

Volumetric Visualization of the Genome

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ABSTRACT

Within the nucleus of a cell, the genetic material (DNA) is condensed, supercoiled, and wrapped around proteins called histones, so to form chromatin. Unlike its traditional visualization, the organization of chromatin is 3D and can be measured with whole-genome sequencing technologies such as Hi-C sequencing, the results of which are often graphically represented as flat 2D heat-maps or a 1D curve that looks like a hairball. Here we discuss a novel approach to represent these and other genomic data by transforming the visualization of the genome from a 1D curve embedded in 3D into a structured 3D volume. This abstraction prioritizes 3D spatial proximity of chromatin and enables advanced techniques like clipping, thresholding, and contouring. Our method allows for multi-variable encoding, thereby enhancing our ability to interpret multivariate relationships, offering a more complete and effective representation of genomic structures, chromosome organization, and data relationships.

Index Terms: Genome, volume, DNA, 3D Hi-C, volumetric, abstraction.

1 INTRODUCTION

In parallel with the advent of genomic sequencing technologies, visualization tools and techniques have been conceived to understand the shape of DNA, compaction of chromatin, and the architecture of the genome. Goodstadt and Marti-Renom review the progress and challenges of genomic visualization over time from a biologist's perspective [1]. They emphasize that advancements in biological knowledge depend on visualization while also recognizing the associated challenges. One of the difficulties stems from the discrepancy between the actual structure of the human genome (very long, extremely thin, intrinsically packed, and dynamic [2]) and the types of objects that the human visual system can efficiently grasp (simple, regular, symmetric shapes [3–6]). Even low resolution models of the genome can be difficult to interpret: as Goodstadt and Marti-Renom state, “a fibrous hairball” does not adequately describe “the overlapping functional landscapes and transitions of state of the genome.”

Nusrat et al. survey the topic of genomics visualization from the perspective of the visualization community, mentioning “the size and complexity of the data” as a potential reason for the limited benefit of the visualization community on genomic data visualization [7]. Genomic data visualization is, however, crucial for interpreting information, generating hypotheses, and communicating discoveries. It links data and algorithmic methods with researchers' cognitive skills. Simple visualization is an increasingly pressing need in the data-driven field of genomics, where many studies lack clear hypotheses [7]. The central challenge is to detect unexpected patterns and form hypotheses from vast genomic datasets. To address this challenge, here we demonstrate an abstraction of genomic organization by means of a simpler, more regular volume. This volumetric approach helps us understand dynamics of 3D organization that were not previously discernible in 3D space.

2 BACKGROUND

DNA is the biomolecule in all of our cells responsible for storing the instructions on how to create different cellular components. If it were laid out end to end it would span nearly 3 meters [8]. However, the size of an average human cell is on the order of tens of micrometers. As such, how DNA is able to fit into a tiny space in the cell remains a complex question. In reality, DNA is rarely linear but is instead elegantly spooled around proteins (histones) and forced into complex 3D “hairballs” by structural machinery [9]. In turn, this compaction of genetic material affects the accessibility of the cell machinery to the underlying genetic material. An outstanding question in the field is whether DNA packaging is a reflection of the needs of the cell to access DNA (function to structure), or whether the compaction itself drives access to genetic material and in turn cell function (structure to function).

3 RELATED WORK

Summaries of genome visualization from the perspective of the biologist are given by Goodstadt and Marti-Renom [1, 10], and from the perspective of the visualization scientist by Nusrat et al. [7].

Because the genome is a 1D chain of nucleotides, the most common visualizations display it along a line or circle. In high resolution only a small portion fits on a screen, and neglects interactions between regions that, despite being far away along the 1D curve, lie close to each other in 3D space [11–13]. Hi-C enables the extraction of the spatial proximities of the genome, and the mutual interactions between all nucleotides are typically visualized through a heat-map matrix [14–16].

Only recently have researchers started to develop 3D models from the mutual distances, which is typically visualized as a colored chain of spheres or a tube [2, 17–20], similar to Fig. 1(a),(b). To our knowledge, the only work showing a volumetric visualizations of the genome is by Lappala et al. [2]. They rendered a tube to represent DNA, with color-coded sections and visualize cross sections of different genomic regions. However, with this approach, most points are covered by multiple tubes/pieces of DNA and the color finally assigned is not deterministic. We present a careful way to address these ambiguities which is general enough to allow the application of other 3D visualization metaphors, which are important to understand the genome [21, 22].

4 DATA

The visualizations are of data previously published by Venu et al., together with the specifics of the data types, experimental procedure, and biological interpretation [23]. They study the dynamic structure-function relationships within the mammalian genome in response to viral infection. The data was generated from experimental infection of Vero cells with the Modified Vaccinia Virus Ankara (MVA) strain. Samples were analyzed at multiple time points to monitor dynamic changes in chromatin structure and function. Two primary methodologies generate the data:

1. Hi-C (Chromosome Conformation Capture Sequencing)—Used to assess 3D chromatin architecture by detecting genome-wide chromatin interactions, and
2. ATAC-seq (Assay for Transposase-Accessible Chromatin Sequencing)—Applied to map open chromatin regions and determine chromatin accessibility.

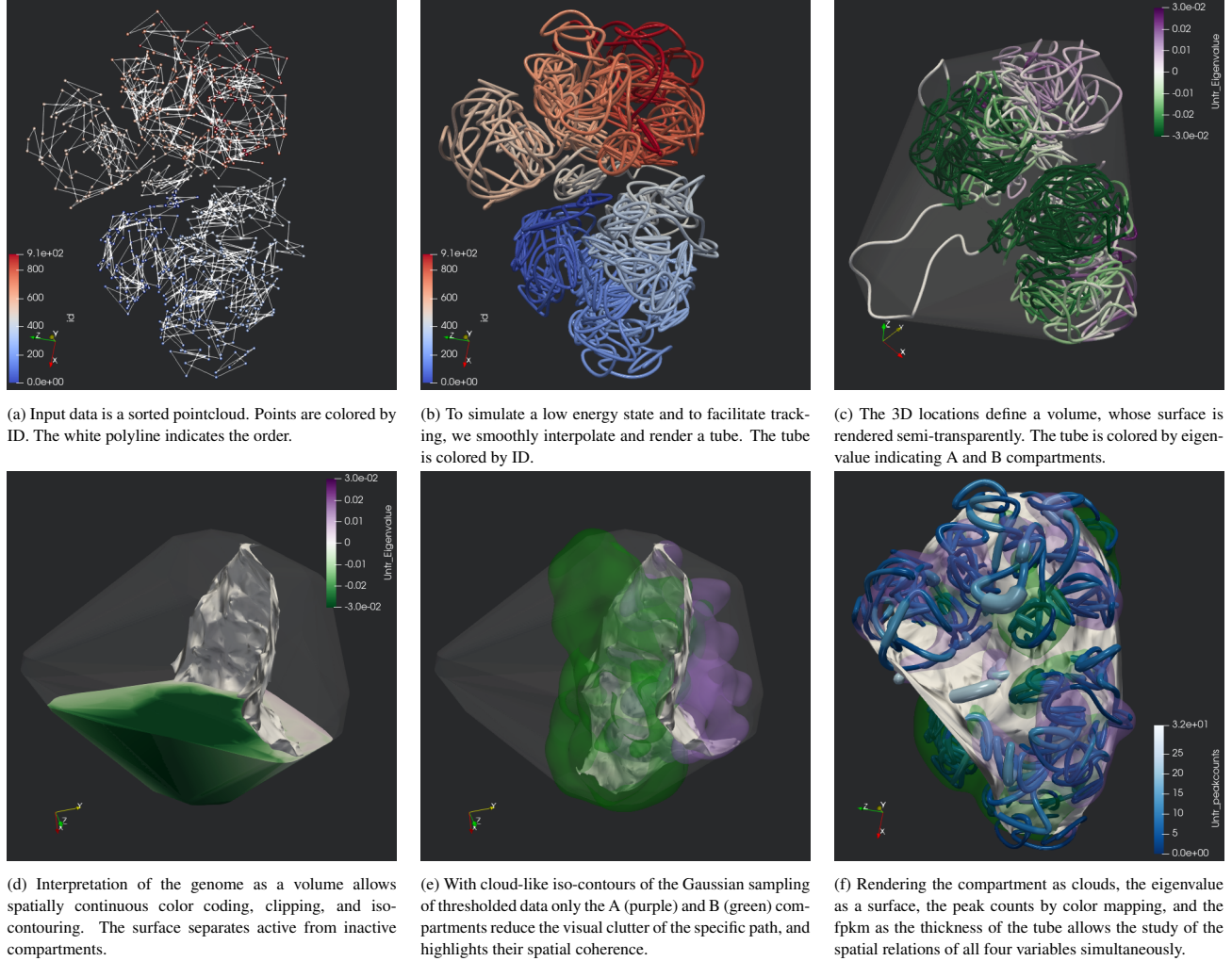


Figure 1: Volumetric interpretation of the 1D reduces clutter, emphasizes spatial coherence, and allows us to apply many different visual metaphors. All of this can be used to generate informative visualizations. This figure shows mock-infected chromosome 4.

For each method, control (mock-infected) and infected samples were processed in parallel. The raw sequencing data is publicly available in the SRA repository under the BioProject accession number PRJNA1037174, with processed data stored in the GEO repository (GSE248052).

Other variables are calculated from ATAC-seq and Hi-C data. They include peak counts and assignments of chromatin compartments. From ATAC-seq data, the number of loci with a significantly enriched sequencing signal are identified [23]. These are counted along a chromosome together with the number of sequenced DNA fragments per kilobase per million (fpkm). From Hi-C, the first eigenvalue is taken from PCA on each matrix; positive and negative eigenvalues are mapped to open (A) and closed (B) compartments, which map to active and closed chromatin states. ATAC-seq data are correlated with these to insure biological interpretability [23].

Recently, Lappala et al. have developed a method to build 3D models from Hi-C data [2]. This method models a chromosome as a continuous polymer, informing its structure via Hi-C contacts to build constraints along the model. This method generates a 3D point set sorted by identifier. Each point represents a region of the genome, which is around 100k to 250k base pairs in our application. This resolution is a tradeoff between fidelity and computational fea-

sibility, where the computation of the 3D structure from the Hi-C data is the bottleneck.

5 VOLUMETRIC VISUALIZATION

We argue that the radical abstraction of genomics data from a one-dimensional curve embedded in 3D to a three-dimensional volume is a valuable approach for gaining insights into genomic structure and function. We demonstrate this in two visualization scenarios that are typical in epigenomics. They involve understanding the interplay of different variables and the influence of viral infection compared to a mock-infected reference. The researchers have combined the results of three experiments to investigate the relationship between structural and functional changes the genome undergoes during viral infection. It is important for them to visualize all experimental data simultaneously. To meet their needs, we have built a visualization that interprets the originally one-dimensional but highly tangled curve as a volumetric object (Fig. 1). This mapping has three main advantages:

1. It reduces the visual clutter from a very long, thin, and tangled object, making the visualization easier to interpret,
2. It emphasizes 3D spatial proximity over 1D proximity, which is crucial for studying topologically associating domains, and

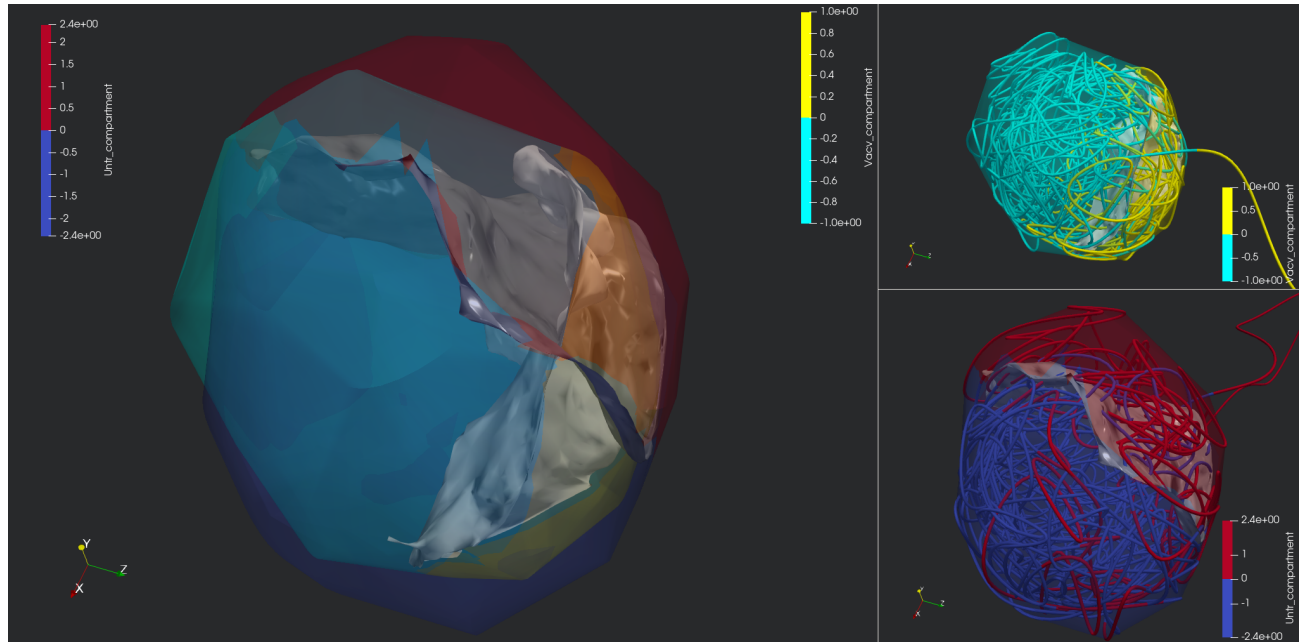


Figure 2: Comparative view of the compartmentalization of chromosome 27 at 24 hours after infection. **Top Right:** The chromatin model (tubes) and chromatin compartments from MVA infected cells, color-coded yellow and cyan for active and inactive compartments, respectively. **Bottom Right:** The chromatin model (tubes) and chromatin compartments from mock-infected cells, color-coded in red and blue for active and inactive compartments, respectively. The inner boundaries of compartments are visualized with a neutral iso-surface. **Left:** The overlaid visualization of chromatin compartments between MVA and mock-infected models.

3. It enables the use of additional processing techniques and visual metaphors, such as clipping, thresholding, volume rendering, and iso-contouring, making possible the encoding of multiple variables in a single view.

We use two different methods to generate volumetric models from the 1D curve. The first applies Delaunay tessellation [24] to compute a tetrahedralization of the convex hull around the spline's points. The second employs Gaussian sampling to produce a continuous 3D scalar density field [25]. Delaunay tessellation partitions points into non-overlapping simplices (triangles in 2D, tetrahedra in 3D) such that no point lies inside any simplex's circumcircle [24]. It avoids skinny elements by maximizing minimum angles, improving numerical stability. Efficient algorithms (e.g., Bowyer-Watson) run in $\Theta(n \log n)$ time, making it scalable [26, 27], and optimized with data structures like quad- or half-edge. Gaussian sampling generates a continuous 3D scalar density field by placing Gaussian kernels at discrete data points, each contributing to the field based on its location and influence radius. The scalar value at any position is computed as the weighted sum of these Gaussians, where weights depend on distance via an exponential decay function. This results in a smooth, spatially continuous representation, useful for interpolation, noise reduction, and visualization, as it captures underlying structures without abrupt transitions. The method is widely used in density estimation, volumetric rendering, and spatial modeling [25]. We find that, compared to Gaussian sampling, Delaunay tessellation can produce sharper edges and corners but yields more accurate hulls and iso-contours (Fig. 1(c)–(e)). The resulting contours do not extend beyond the data's bounds, thus avoiding biologically irrelevant closed surfaces. We have applied these techniques in two scenarios. In the first, the researchers aim to understand spatial relationships among variables from different experiments. In the second, they want to compare how a mock-infected cell and a virus-infected cell evolve over time.

5.1 Comparing Variables

The input to our visualization pipeline is a 3D point set representing the 1D curve and associated with several scalar variables. An `id` array identifies the genome's path (Fig. 1(a)). To facilitate analysis, we use a smooth interpolation technique to transform discrete points into a continuous representation. Smooth splines are typical in genome visualization because they make the 3D path easier to follow and mimic a low-energy configuration (sharp bends represent stress), as shown in Fig. 1(b). We then render this spline as a tube. Although its thickness is greatly exaggerated compared to the genome's actual 2 nm diameter, this trade-off is necessary for visibility. Other benefits include color-coding or thickness-coding of variables and improved depth perception through shading.

We found it useful to combine both volumetric representations, i.e., Delaunay and Gaussian, in a single visualization. Because they behave so differently, they are easy to distinguish, allowing multiple layers of information to be integrated. In Fig. 1(f), we show the final visualization pipeline, displaying eigenvalue, compartmentalization, peak counts, and fpkm in one image. This enables researchers to examine their relationships with high fidelity [28].

5.2 Comparing Infection Types

In this case, we have two time-dependent datasets: one from a vaccinia-infected cell and one from a mock-infected reference. Each dataset is averaged over several replicates. To maximize visual comparability, we align them by translation and rotation. We do not rescale them, since infection may cause the genome to expand or contract, which is of interest to researchers. For translation, we match the centers of mass for each dataset; for rotation, we use the product of the left and right eigenvector matrices from the singular value decomposition of the covariance matrix as described by Umeyama [29], which is mathematically proven to produce the optimal result with minimal average distance between the corresponding points.

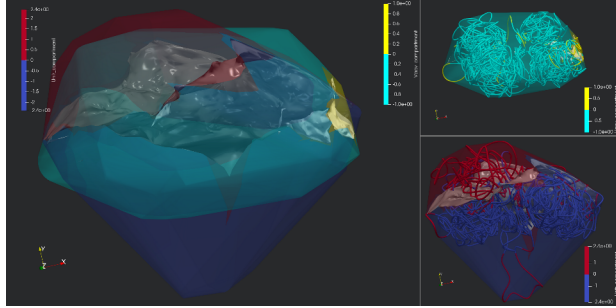


Figure 3: Visualization as in Fig. 2 for chromosome 4.

Overlaying both genomes can introduce visual clutter and overwhelm researchers [28]. To reduce cognitive load, we provide an overview by displaying only a single binary variable—compartmentalization—by coloring the smoothed convex hull surface at the boundary between compartments A and B. For detail on demand, we also include two smaller juxtaposed views, each with the tube rendered for context, and link the camera in all views [30]. For the color scheme, we chose red/blue and yellow/cyan for high perceptual contrast and minimal confusion when colors overlap. We also wanted intuitive associations [31, 32]: red and yellow suggest reactive “fire” for active compartments, and blue and cyan suggest calm “water” for inactive compartments. Finally, we ensured that the perceptual distance between A and B compartments within each dataset is similar and that their luminances are similar [33].

6 EVALUATION

We have collected feedback from six domain experts. Our collaborating biologists value the new possibilities they are given from the visualizations. Referring to Venu et al. [23], Dr. Steadman states: “If we had had this technology earlier, we would have written a completely different paper. The genome must be looked at in 3D.”

Within the cellular nucleus, chromosomes form large domains of active and inactive states, known to the Hi-C community as the open (A) and closed (B) chromatin compartments, respectively [34]. The assignment of these chromatin states can be estimated directly from the Hi-C data [35]. Alterations in chromatin A/B compartment assignments are easy to detect yet difficult to interpret. For example, Venu et al. calculated the compartment scores for both MVA and mock-infected samples, regressing these values against each other to determine the extent of compartment reassignment (for example, see Fig. 2D from Venu et al. [23]). This analysis identified regions of chromatin that either switch assignments (from A to B, or B to A) or maintain their original assignment post-infection. However, previous analysis did not show where compartment switching takes place, how compartments are maintained along a chromosome, or the auto-correlation of neighboring assignments in 3D space. Referring to their previous analysis in Venu et al. [23], Dr. Roth states, “Normally, we view the compartment assignments as a 1D signal track that oscillates in value between open and closed states [along a linear genome]. These [volumetric] figures help us see how different portions of a chromosome share a common compartment assignment and shift between states post-infection, which we did not see before.” With new volumetric visualizations, understanding changes to chromosome compartments is simpler. In Fig. 2, we can clearly see the overlap in stationary compartments (orange region painted along chromosome 27) that maintain their compartment assignment across infection. We can also appreciate that chromosome compartment switching is better described as compartment shifting along the outer surface of the chromosome.

For example, the biologists had clearly seen the strong influence

of the infection on chromosome 4, Fig. 3, where the active compartment is significantly reduced but they had thought that chromosome 27, Fig. 2, had remained almost unchanged. Only now had it become apparent that, despite the size of the active compartment remaining approximately constant, the spatial location had significantly changed.

We want to emphasize that there are cases where the convex hull is more suitable for creating a clear visualization, and others where the Gaussian cloud is preferable. Using the surface of the convex hull, as shown in Fig. 1 (c)–(e), significantly simplifies the genome’s shape because all concave structures are lost. Application scientists have specifically requested a semi-transparent convex hull overlay in the visualizations, as it provides a frame of reference for the overall spatial extent of their data without cluttering the image with shading details of concave regions. In contrast, the Gaussian cloud metaphor addresses this limitation by preserving more structural detail. The Gaussian representation is especially valuable for detailed shape analysis, as in Fig. 1 (e) and (f), where a specific variable is the focus. However, presenting this level of detail consistently would overwhelm the application scientists’ visual system. We have experimented with overlays similar to those in Fig. 2, replacing the convex hull with Gaussian clouds. As a team, however, we had to reject these overlays because the images became too cluttered to mentally separate the two infection states.

7 CONCLUSION

Effective visualization of genomic data is essential for interpretation, hypothesis generation, and communication. A key challenge lies in reconciling the genome’s actual structure—long, and highly compacted and entangled—with forms that are more perceptually intuitive, such as simple, regular, and symmetric shapes. In this work, we address this through abstraction, transforming the genome from a 1D curve embedded in 3D into a 3D volume. To our knowledge, this is the first approach that replaces the tangled-hairball representation with a more structured volumetric form. This transformation reduces visual clutter, making the data easier to interpret. It shifts the emphasis from 1D to 3D spatial proximity, which is critical for studying genomic interactions. Additionally, it enables the use of advanced visualization techniques such as clipping, thresholding, volume rendering, and iso-contouring, allowing multiple variables to be encoded within a single view. Through collaboration with domain experts, we demonstrate that this approach enhances clarity and facilitates more effective exploration of complex genomic structures.

Our visualization uses abstraction to translate complex phenomena into a form that is familiar to an observer. This allows the development of intuition about qualitative properties and changes in the genome. These intuitions are not taken as truths by the subject matter experts, who understand the tradeoff between simplicity and accuracy in a visualization. The intuitions’ purpose is to inspire the comparison with lower-dimensional flattened visualizations and the design of quantitative analyses from statistics to validate or disprove the intuitive interpretation. Our collaboration has shown the power of abstract visualizations to spark ideas, and has resulted in meaningful questions that the researchers would simply not have asked without it. The potential danger, that the intuition can be wrong because abstraction and simplification can hide details and shift the true nature of the data, has not been problematic because the researchers generally do not take the visualization as the final scientific result, but use it as a means to generate hypotheses and design rigorous analyses and experiments.

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